ENPH 455 Final Report

FitsQuery: A Search Tool for Imaging Surveys

by

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Abstract/Executive Summary

The KFPA Examinations of Young Stellar Object Natal Environments (KEYSTONE) survey mapped ammonia emission across 11 Giant Molecular Clouds (GMC) and stored this data as Flexible Image Transport System (FITS) files on the Canadian Advanced Network for Astronomical Research (CANFAR) platform. FITS files are the preferred storage format for astronomical data and are thus the most commonly used file type across astronomical platforms and catalogues. The KEYSTONE files stored on the CANFAR platform, including other archives and catalogues, are able to be queried based on region or object type. This can be done by converting the world coordinates to pixel coordinates and matching the values to NAXISn or the number of axes in the data array.

In this thesis a new database (FitsQuery) was designed to query FITS files for a given region and object type and verify whether the area of interest is located in the target survey or catalogue. The primary survey that FitsQuery was designed around was the KEYSTONE survey but can be adapted to target other surveys or catalogues. It is a tool that is capable of alleviating the slow task of manually downloading files and seeing if it contains the region of interest. It is also capable of providing snapshot images of that area. These include the Moment maps, temperature, mean and median over the area, and average spectrum. However, given that the program was originally designed around the KEYSTONE survey, some of images may be tailored to mapping the data for ammonia emissions or the KEYSTONE data cubes, which cannot be universally applied to other datasets. Further expansion of this database should include continuing to expand the options available while also ensuring that they can be universally applied to common forms of astronomical data and easily altered to do so.

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1 Introduction

Astronomical databases and archives are helpful platforms for astronomers to easily access data collections and surveys. One such database is SIMBAD or Set of Identifications, Measurements, and Bibliography for Astronomical Data that is maintained by the Centre de Données astronomiques de Strasbourg (CDS) [1]. It contains approximately 13,000,000 astronomical objects, excluding Solar System bodies, in the database and can query either internally or externally, such as through web access or in Python [2][3]. Another database is the National Space Science Data Center (NSSDC) that contains various data sets, including data from the Cosmic Background Explorer (COBE) [1]. This data set is contained in the NSSDC in the standard Flexible Image Transport System (FITS) [1] [4].

The FITS file was first proposed as a means for astronomers to encode the definition of the astronomical data and the data or the Header and Data Unit (HDU) [1] [4]. The former part of the FITS is known as the Header Unit, which describes or defines the data as a series of keywords, and the latter is the single data array or the Data Unit, which contains the data described in the Header [4]. FITS or FITS files are the formalized means of mapping data arrays to a physical coordinate system and the format commonly used by platforms and archives [4].

The Canadian Advanced Network for Astronomical Research (CANFAR) is a platform that easily distributes and stores massive astronomical data sets for users to then process and examine [5]. Currently it hosts approximately 22 diverse major collections, such as James Clerk Maxwell Telescope (JCMT), Canadian Galactic Plane Survey (CGPS), and Far Ultraviolet Space Explorer (FUSE) [5][6]. Another data set is the KFPA Examinations of Young Stellar Object Natal Environments (KEYSTONE) survey that uses the Green Bank Telescope (GBT) to map NH_3 emission over eleven giant molecular clouds (GMCs), including their median cloud kinetic temperature, centroid velocity, density, and velocity dispersion [7].

The new search tool FitsQuery was developed using the KEYSTONE data that is stored and accessed through CANFAR [5]. The intention is to allow for astronomers and physicists to have a quick and easy way to verify if the region or object exists in a targeted collection or survey. The next is to subsequently provide various types of snapshot images of that region.

This alleviates the need for astronomers to manually search through files by confirming if that specified region exists in the files while also displaying images of that region.

1.2 Motivation

There are existing databases and libraries that query, display, edit, and analyze astronomical datasets. One such software is Cube Analysis and Rendering Tool for Astronomy (CARTA) [8]. This downloadable tool allows for users to visualize and analyze large downloaded data files, including FITS and HDF5 [8][9]. Astro-FLEX or Astro-Fits Lite Explorer and ASTAP or Astrometric STAcking Program are other software and tools that can be used to load, examine, and view FITS [10] [11]. However, many of these tools and software are used to either display or query astronomical data [12]. This program is intended to replace the need to continually download and query massive files by being able to both query and visualize FITS files in any given database. Furthermore, the process of downloading and scanning various files within a dataset to locate a specific region can be time consuming and tedious, especially if the target does not exist within those files. This is in addition to, if an astronomer were to open a file, search, and then discover it does not have the targeted area, other files would then have to be downloaded and queried. Downloading such large files can require a large amount of storage space and can waste and occupy a significant amount of data storage on a hard drive [13]. Overtime these large, downloaded files can affect the performance of a computer by slowing down its processing speed and thereby increase the amount of time it takes to download other files [13]. Therefore, a method of directly accessing and querying FITS directly would save astronomers and researchers from downloading unnecessary files and prevent storing such files on their hard drive.

1.3 Problem Statement

A new search tool for imaging surveys was designed to easily query FITS files from a database or survey. It was designed so that a user simply inputs either an object name/type or celestial coordinates and it will tell the user if it is found within those files. FitsQuery additionally provides prompts to better target a certain region and provide a quick snapshot of that area. Such requests consist of selecting a particular plane and viewing the Moment maps [7]. The program was designed and tested using the KEYSTONE survey data that is accessed

using CANFAR. It was ensured that any astronomer or researcher can easily change the target catalogue or data set, such as change KEYSTONE to ALMA. The database has no adverse environmental effects or sustainability issues, no need for ethical or equity considerations, and no safety risks. All data and software used is publicly available and is easy to implement so that anyone can access and run the database. Potential constraints for the project is the time it could take to query and process each file. Each file differs in size and therefore some files may take longer to process than others. Thus, there is the potential that too long of a processing time could make the program redundant. This constraint was eased by ensuring the optimization of the code.

1.4 Methodology

The program language chosen for this database was Python. This allows for other existing astronomical libraries and packages to be implemented. Astropy is a popular Python package that allows users to support and analyze specific astronomical file formats, including FITS [14]. It was employed in a variety of ways throughout FitsQuery, such as converting pixel coordinates to world coordinates. Astroquery is another Python package that can be used to query archives and catalogues, such as ALMA. It can be used to query other datasets when FitsQuery is altered to target other archives.

2 Background

2.1 Celestial Coordinate System

Where Earth has terrestrial coordinates that are defined as latitude and longitude, the celestial sphere or the sky is defined with a coordinate system known as the *equatorial* system [15]. In this system, the two coordinates or celestial coordinates are right ascension (RA) and declination (Dec) that are roughly fixed for each star; right ascension is the equivalent to the terrestrial longitude and declination to the terrestrial latitude [15]. A spherical representation of this coordinate system can be seen in **Figure 1** where δ is the declination, α is the right ascension, γ is the fixed point on the celestial equator (vernal equinox), and the north and south poles are now the north and south celestial poles [15],[16],[17].

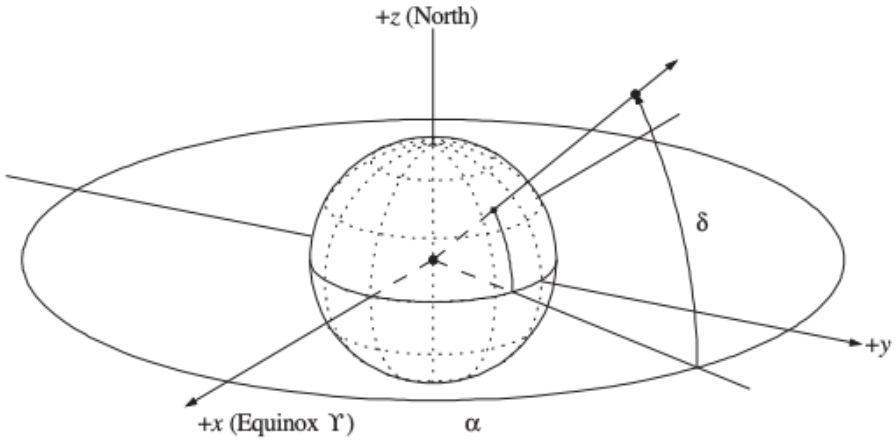


***Figure 1**: Representation of the celestial sphere with the right ascension (α) and the declination (δ) labelled on their respective axes. Image was taken from [18].*

Celestial coordinates are represented in the FITS header and can be used to query a file by inputting the right ascension and declination. As seen in the breakdown of a FITS file in **Figure 2**, each file contains data for specified pixel coordinates or CPRIXn [19][4]. The physical coordinate type is defined in CTYPEn while CRVALn is the coordinate value at that reference pixel; in the example outlined in **Figure 2**, the right ascension would be 309.553° and the declination would be 42.0323° [20].

Declination is measured in degrees from the north or south of the celestial equator [17]. Right ascension is measured towards the east on the celestial equator from γ to intersect the hour circle of an object [17]. While declination is in degrees, right ascension is typically measured in terms of hours, minutes, and seconds where 24 hours is equal to 360° [17]. The relations between the two coordinate conventions can be seen below [15]:

$$1^h = 15°, 1^m = 15', 1^s = 15''$$

$$1° = 4^m, 1' = 4^s, 1'' = \frac{1}{15}^s$$

Celestial coordinates can be formatted as right ascension 00h00m00s and declination $00°00'00''$ or typically formatted in the more common sexagesimal 00:00:00 [21].

```
SIMPLE = T / conforms to FITS standard
BITPIX = -64 / array data type
NAXIS = 2 / number of array dimensions
NAXIS1 = 566
NAXIS2 = 776
WCSAXES = 2 / Number of coordinate axes
CRPIX1 = 283 / Pixel coordinate of reference point
CRPIX2 = 388 / Pixel coordinate of reference point
CDELT1 = -2.444048104767E-03 / [deg] Coordinate increment at reference point
CDELT2 = 2.444048104767E-03 / [deg] Coordinate increment at reference point
CUNIT1 = deg / Units of coordinate increment and value
CUNIT2 = deg / Units of coordinate increment and value
CTYPE1 = RA---TAN / Right ascension, gnomonic projection
CTYPE2 = DEC--TAN / Declination, gnomonic projection
CRVAL1 = 3.095539691573E+02 / [deg] Coordinate value at reference point
CRVAL2 = 4.203231621938E+01 / [deg] Coordinate value at reference point
LONPOLE = 1.800000000000E+02 / [deg] Native longitude of celestial pole
LATPOLE = 4.203231621938E+01 / [deg] Native latitude of celestial pole
RESTFRQ = 2.369450000000E+10 / [Hz] Line rest frequency
RADESYS = FK5 / Equatorial coordinate system
EQUINOX = 2000 / [yr] Equinox of equatorial coordinates
SPECSYS = LSRK / Reference frame of spectral coordinates
SSYSOBS = TOPOCENT / Reference frame of spectral observation
BUNIT = km.s-1
BPA = 0.000000000000E+00
BMIN = 8.554168366684E-03
BMAJ = 8.554168366684E-03
```

***Figure 2**: Fits header extracted from CARTA for W3_west_NH3_11_all_rebase_multi_mom0_QA.fits.*

2.2 Spectral Lines

Giant Molecular Clouds (GMCs) are huge complexes comprised of interstellar dust and gas that are left over from the formation of the galaxies [22] These huge accumulations of gas make up most of the interstellar medium (ISM) [23]. For GMCs, temperatures typically range from 10 to 20 K, number densities 100 to 300 cm^{-3}, and have a typical size of approximately 50 pc and can be distinguished by having a mass exceeding 10^6 $M_\odot$ [17]. They are primarily composed of H_2 with the second most abundant molecule being CO [23]. This allows for the emission an absorption to be observed. Despite H_2 being the primary composition, the molecules do not have any emission or absorption lines in the visible area of the

electromagnetic spectrum making them difficult to be observed [17]. This leads to other molecules being used as tracers with CO being the most commonly used (**Figure 3**); other molecules include CH, OH, CS, and C_3H_2 [17]. These tracer molecules become excited or de-excited during collision causing spontaneous transitions between the excited states resulting in spectral lines [17]. This causes photons to be emitted in wavelength regions that can be more easily observed [17].

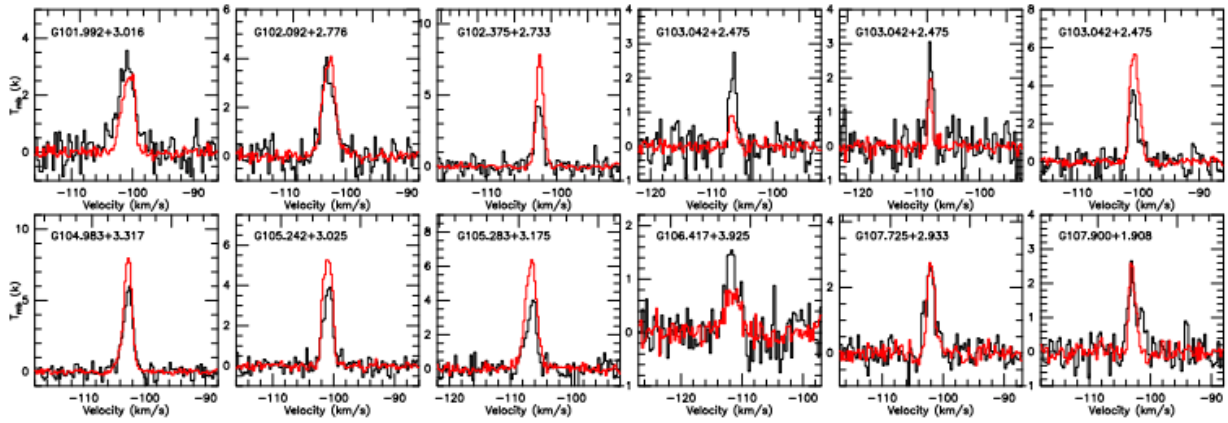


***Figure 3**: Observed CO spectra towards the Galactic edge clouds extracted from the CO (1-0) emission peaks. Image is from a study of molecular clouds at the edge of the galaxy [24].*

Molecular clouds have currently been observed by means of stellar extinction due to dust, the thermal emission of dust, and the rotational and vibrational spectral lines of the molecules [25]. Another methods is using the abundance of dust and its optical thinness as a tracer by analyzing changes in stellar reddening that is caused by the dust to estimate the distance to the cloud [26]. Molecule tracers from gas (e.g NH_3 and N_2H^+), however, can provide other advantageous over the more common carbon-based molecule tracers [7]. These dense gas emissions instead allow for examination of dense core and filament kinetics as they are less affected in high dense cores by freeze-out from dust grains and are optically thin [7]. Another reason can especially be seen in the hyperfine splitting of ammonia which allows for a more convenient method of retrieving optical depths and excitation temperatures [7]. The ammonia (1,1) inversion line is comprised of a main line with two hyperfine lines on either side [27]. The interaction of the hydrogen nuclear spins interacting magnetically with the total angular momentum of the molecule cause these hyperfine structures in spectral lines that are in the inversion spectrum [28]. The inversion lines for NH_3 make it a desirable tracer when studying

molecular clouds given its abundance and hyperfine structure [29]. The advantages and use of ammonia can be seen in the KEYSTONE survey.

2.3 Spectral Cube and Moment Maps

A spectral cube contains three-dimensions comprised of both spatial and spectral information [30] [31]. In the FITS header, a spectral cube can be identified if it contains NAXIS3 (number of axes in the array), which is comprised of right ascension (x), declination (y), and a spectral dimension (z axis) [20], [30], [31]. A file of only NAXIS2 is a simple 2D map with no spectral component and only right ascension (x) and declination (y) [30], [31]. These cubes are 3D images of a specific area and pixel in the sky where the first two axes show its position of the galaxy and the third axis shows how the gas is moving in the galaxy [32]. An example of a spectral cube can be seen in **Figure 4**.

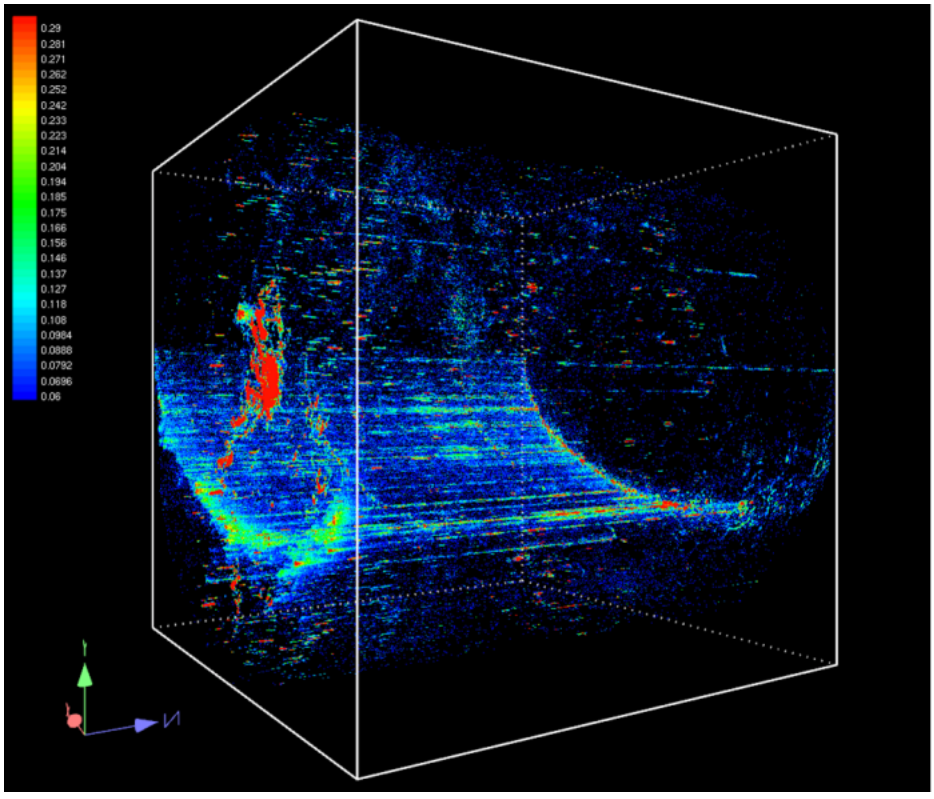


***Figure 4**: Rendering of the HI Parkes All Sky Survey (HIPASS) cube of the entire southern sky generated by Russell Jurek. Image was taken from [33].*

Moment maps can be created using the 3D spectral cubes that help create summary images that are easier and faster to interpret and process [32]. The "Moment" is defined as the

weighted integral along the axis, usually the third spectral axis [31]. Moment 0 maps are the integrated intensity over the spectral line or where the gas is and how much there is (**Figure 5**) [32], [34]. This can be seen in equation 1

$$M_0 = \int I_v dv \tag{1}$$

where I_v is the intensity and v represents the velocity, frequency, energy, or wavelength that is in the spectral axis [32], [34].

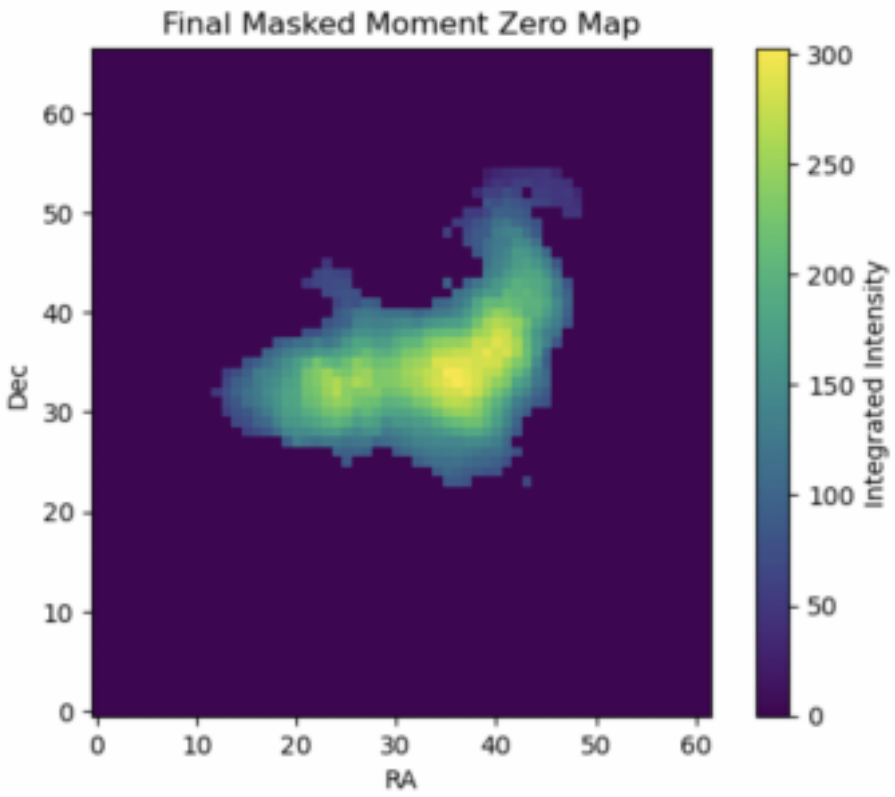


***Figure 5**: Moment 0 map for MRK54. Image taken from [32].*

Moment 1 maps show the intensity-weighted velocity of the spectral line or the how the gas is moving (**Figure 6**) [32], [34]. This can be seen in equation 2

$$M_1 = \frac{\int v I_v dv}{\int I_v dv} = \frac{\int v I_v dv}{M_0} \tag{2}$$

where M_0 is the previously calculated Moment 0 [32], [34].

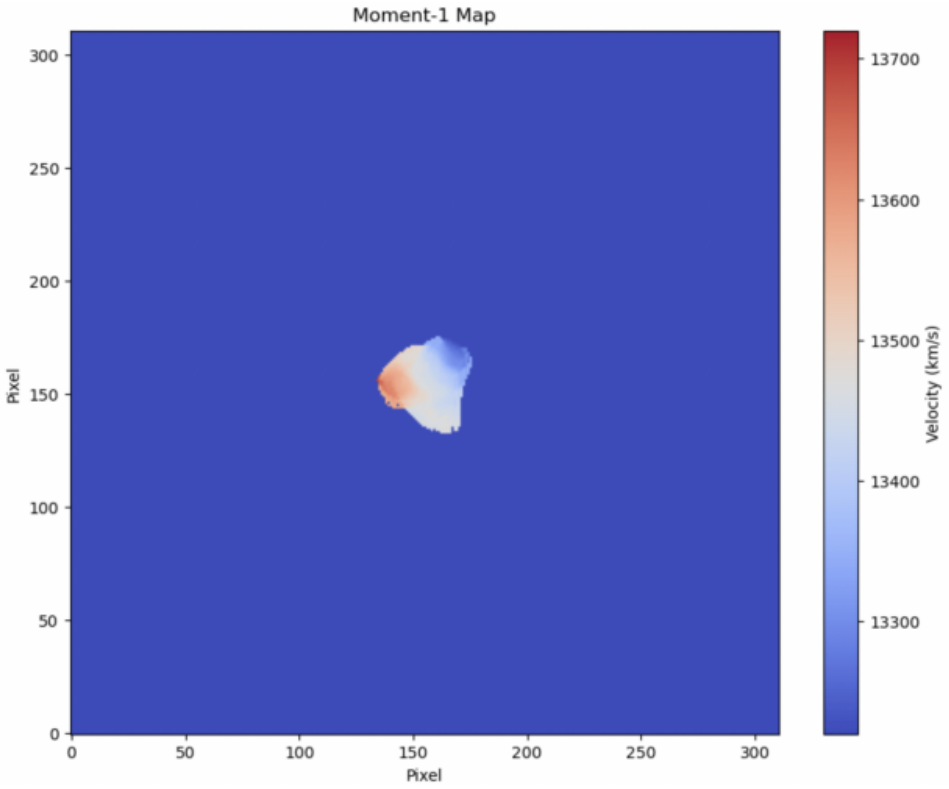


***Figure 6**: Moment 1 map for MRK54. Image taken from [32].*

Moment 2 maps show the velocity dispersion or how turbulent the gas is (**Figure 7**) [32], [34]. This can be seen in equation 3

$$M_2 = \frac{\int vI_v(v - M_1)^2 dv}{M_0} \tag{3}$$

where M_0 and M_1 are the previously calculated Moment 0 and Moment 1 [32], [34].

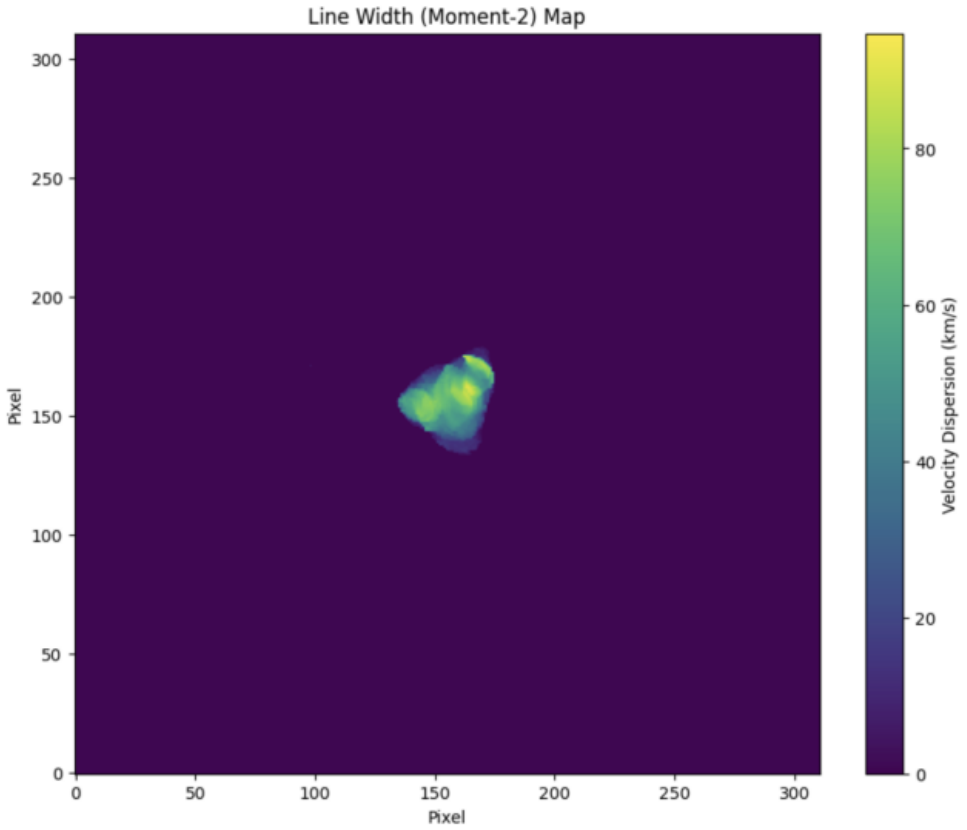


***Figure** 7: Moment 2 map for MRK54. Image taken from [32].*

It is typical to splice the cube and cut along the spectral axis or select a plane to have a 2D dataset or image [31].

3 Results and Discussion

The initial version of FitsQuery queried the KEYSTONE files only by matching the celestial coordinates and outputting the number of all known matches. Once it was confirmed that the program was efficient in searching the header and locating the NAXISn, it was no longer necessary to continue searching through all files once a match was detected. The code was then changed so that, when a match is detected, the program stops iterating through all the files and announces to the user that a file has been found; if no file is found, it will iterate through all files before notifying the user that there is no file match. After various research and iterations, it was found that the FITS files for KEYSTONE are only able to be accessed through their URLs and not through implementing Astroquery. The KEYSTONE URLs are saved in a separate text file. FitsQuery has a function that opens this text file and strips any

leading characters [36]. The code is able to be adapted by altering lines of FitsQuery. The ALMA archive and Dataverse have both been successfully integrated into the code and tested to see if it can query using known coordinates (**Appendix B. Alternative Catalogues**). Currently, it has only been successfully tested with querying a file and detecting if the coordinate is found or not found.

Once a file has been matched to the coordinates, it outputs available information, such as the number of axes and spectral planes, before asking the user to specify a spectral plane. If they select yes, they will be asked to choose from the available planes [31]. If the user selects no, it will be treated as a spectral cube [31]. The chosen plane will be used in the subsequent images. Next the user is prompted to select a circular or rectangular region shape before being asked to enter either the radius or width and height in arcseconds. Finally, once these parameters have been established by the user, a set of options of image types will be shown to the user whereby they can continue to select a type of image, such as Moment 0 map for integrated intensity, and the option to save the image. A flowchart can be seen in **Figure 8** that shows how the current version of FitsQuery iterates and runs through each prompt and output. A comprehensive breakdown of the query tools and prompts are described below.

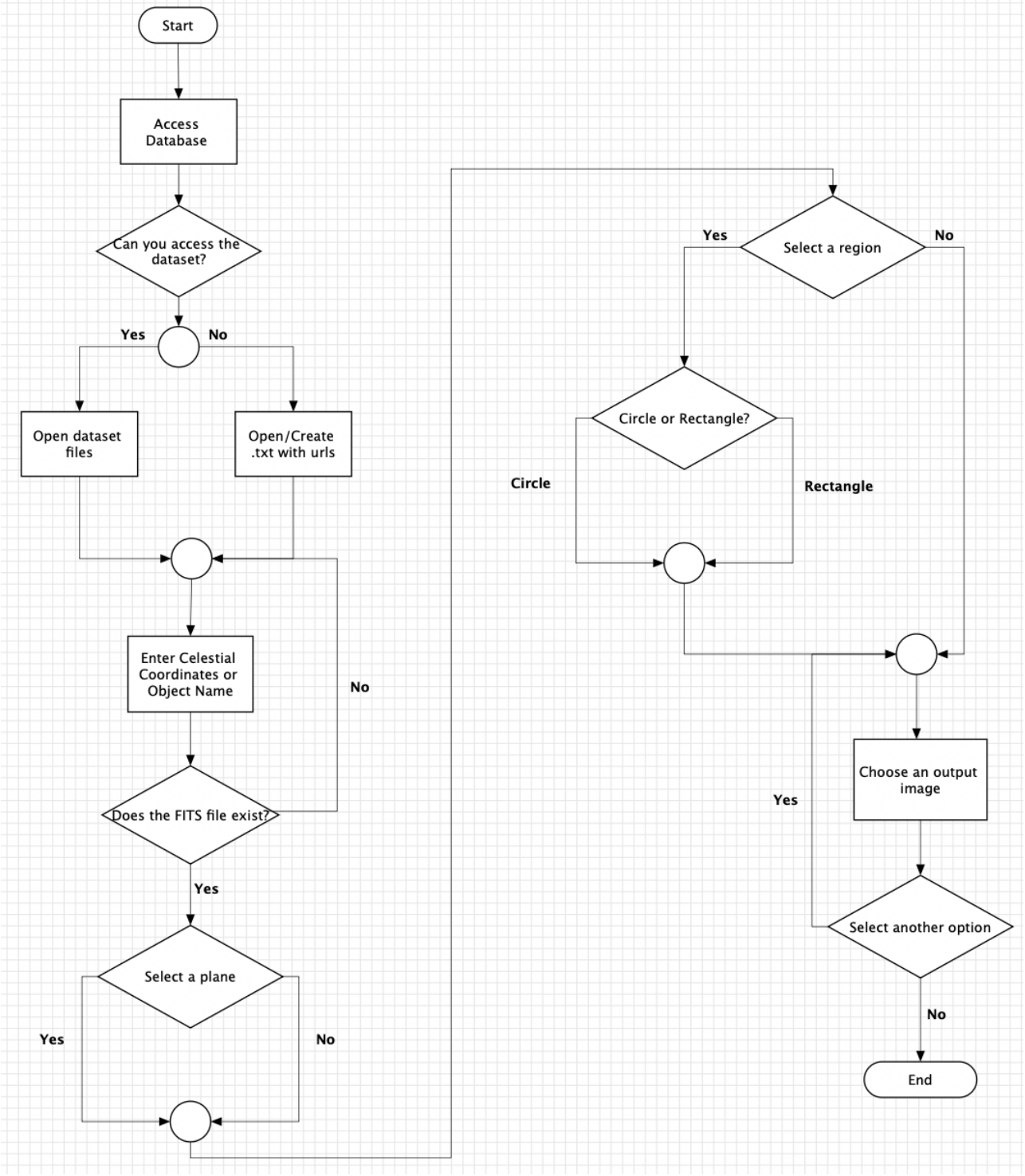


***Figure 8**: Flowchart of the logic for the current version of FitsQuery.*

3.1 Query Tools

The first prompt in FitsQuery allows the user to query by entering either coordinates or object name. This input is taken by the program and compared to the NAXIS1 and NAXIS2 in the HDU. If selection 1 is chosen, the user enters the exact object name. The program takes the name and uses Simbad.query_object from astroquery.simbad to obtain the right ascension and declination. If selection 2 is chosen, the right ascension and declination are already imputed by the user. These world coordinates are then converted to pixel coordinates and matched to the NAXISn. From these coordinates, FitsQuery will be able to locate the first FITS file with the matching set of coordinates. An example of how a user may input a target can be seen in **Figure 9**. FitsQuery was tested to confirm if the program had successfully queried a region located in the KEYSTONE survey by inputting known object names and coordinates. All known object names and coordinates were successfully tested and confirmed. These are found in **Table 1**. Randomized coordinates were used to test if the program successfully detected coordinates that were not found within the files.

***Table 1**: A list of the KEYSTONE Target GMCs [7].*

Region	**RA (J2000)**	**Dec (J2000)**
W3	02:23:22.140	+61:36:17.432
Mon R2	06:08:25.657	−06:14:32.812
Mon R1	06:32:32.294	+10:27:13.335
Rosette	06:33:38.530	+04:29:10.771
NGC 2264	06:40:41.339	+09:25:42.177
M16	18:18:38.140	−13:39:30.050
M17	18:19:35.479	−16:19:09.088
W48	19:00:52.657	+01:41:55.338
Cygnus X South	20:33:42.800	+39:35:41.356
Cygnus X North	20:37:14.998	+41:56:04.742
NGC 7538	23:14:50.333	+61:29:04.744

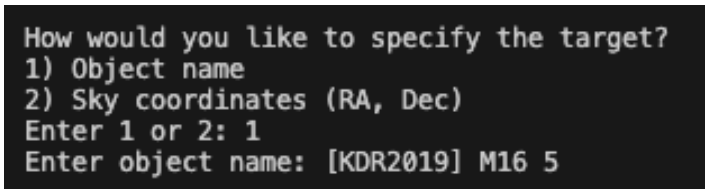


Figure 9*: Screenshot of the prompt that asks the user to specify the target with example input parameters.*

3.1.1 Query by Object Name

FitsQuery can query the sky using object types and locate the matching FITS file. When an object name is inputted, FitsQuery locates the object type in SIMBAD and accesses its sky coordinates in tandem with the coordinates package in astropy. It takes these coordinates and scans through the files. When a match is found, it announces it to the user alongside a breakdown of the available information, such as how many planes. A successful match can be seen with the object type [KDR2019] M16 5 (RA 18:18:20.4 and Dec -13:46:16). The program uses Simbad.query_object() to obtain the celestial coordinates, index them as right ascension and declination, and then saves them as the working coordinates using SkyCoord from astropy [37], [38]. When a match is not found, FitsQuery announces to the user that there has been no match. An unsuccessful match can be seen with the object type Orion (RA 05:35:16.8 and Dec -05:23:15) where “No file contains these coordinates” is printed after attempting to query.

3.1.1 Query by Celestial Coordinates

FitsQuery can also query the sky using the sky coordinates (right ascension and declination) and locate the matching file. When the coordinates are inputted, FitsQuery uses the coordinates package in astropy and scans through the FITS files. The outputs are the same as querying by using object name. A successful match can be seen with a RA 18:18:20.4 and Dec -13:46:16 where “A matching file has been located!” and the coordinates are saved as the working coordinates using SkyCoord from astropy in degrees [37]. The number of axes and spectral planes available are displayed to show the user details about the file. An unsuccessful match can be seen with RA 05:35:16.8 and Dec -05:23:15 where “No file contains these coordinates” is printed after attempting to query.

3.2 Selection Options and Outputs

After a file match is located, there are additional prompts for the user to provide additional criteria before outputting an image. The first is that they may specify a spectral plane. This allows for the user to slice the cube along a particular plane rather than view the entire cube. The next prompt is the user may select a region shape and size for their region of interest. The default option is a circular region with a radius of 60 arcseconds. This criterion allows the user to be more precise in viewing their inputted coordinates. SIMBAD has a similar viewing feature. The user may select a circle or rectangle and provide the corresponding dimensions in arcseconds (**Figure 10**).

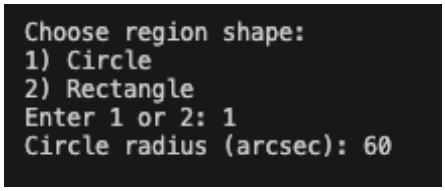


Figure 10*: Screenshot of the prompt that asks the user to input a region shape and size in arcseconds with example input parameters.*

The region is centered around the coordinates that were given in the first prompt. The last prompt is for the user to select which type of image they wish to view. The current options are broken down below. When an image is exited, they have the option to save the image otherwise they may select another image type. The last option is to exit the loop.

3.2.1 Moment Maps

The first three options are the Moment maps. This option allows users to select a particular plane to view from the data. The Moment 0 map shows the NH_3 integrated intensity that is described in equation 1 [7], [34]. The KEYSTONE survey contains files that already contain the Moment 0 map. This allows the program to avoid calculating the integrated intensity from the full spectral cube by simply locating the plane and Moment 0 file. However, the code accounts for if these files do not exist in other databases and, if not Moment 0 file or plane is detected, it instead calculates the Moment 0 using equation 1. A Moment 0 map can be seen in **Figure 11** that is targeted at specific object ([KDR2019] M16 5), which is a dense core [39].

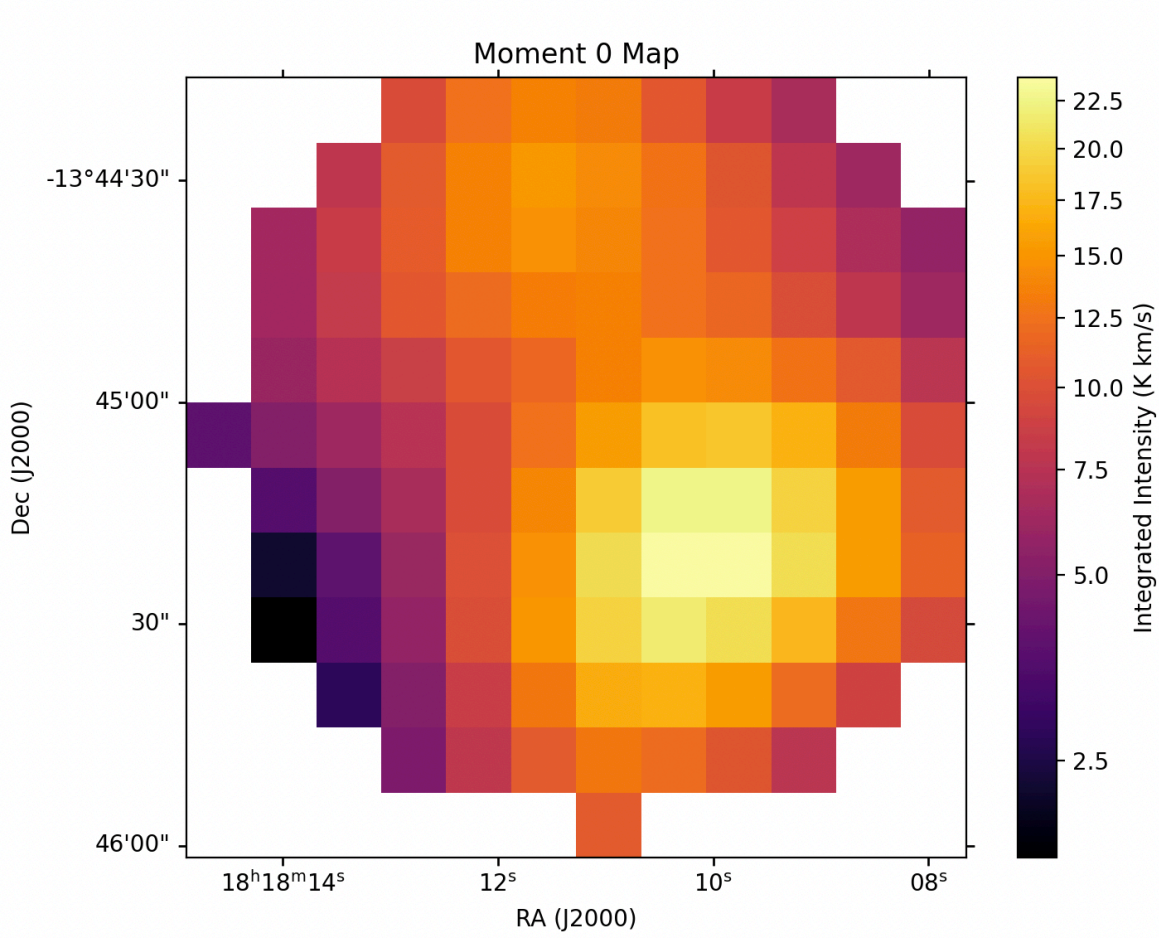


***Figure 11**: Moment 0 map for [KDR2019] M16 5 (RA 18:18:20.4 and Dec -13:46:16) querying around a circular region of 60 arcseconds.*

The Moment 1 map is the next option. The Moment 1 function requires the full spectral cube. The function first transforms the pixel coordinates to world coordinates using the pixel coordinate of the reference point (CPRIX), CDELT, and CRVAL [20]. The linear transformation can be seen in equation 4

$$q_i = \sum_{j=1}^{N} m_{ij}(p_j - r_j) \tag{4}$$

where the p_j is the pixel coordinate, r_j is CPRIX, m_{ij} is the matrix of size NxN, and q_i is the offset intermediate world coordinate that is converted to its corresponding world coordinate, x_i, in equation 5 using s_i (CDELT) [20].

$$x_i = s_i q_i \tag{5}$$

The converted world coordinates, which are known to be frequency in Hz in the spectral plane from CUNIT3 (units for the world coordinate values), are converted to velocity using the doppler shift for spectral lines (equation 6) [17],[4]. The relation of wavelength and frequency (equation 6) can be used to obtain the velocity in terms of frequency (equation 8) [17].

$$\frac{v}{c} = \frac{\lambda_{obs} - \lambda_{rest}}{\lambda_{rest}} \tag{6}$$

$$\lambda = \frac{c}{\nu} \tag{7}$$

$$\frac{v}{c} = \frac{\nu_{rest} - \nu_{obs}}{\nu_{rest}} \tag{8}$$

This is necessary since the KEYSTONE data cube contains frequency (Hz) and not velocity (km/s). The velocity obtained from the spectral frequency is now used to compute Moment 1 (and Moment 0) using equation 2 [34]. When the user selects a Moment map, they must enter the upper and lower velocities to be used as the bounds for the integral. The function generating the Moment 2 map is similar to the Moment 1 map apart from computing the Moment using equation 3. The KEYSTONE survey has extensive and complex maps for each Moment (**Figure 12**), however a map showing the velocity dispersion at the targeted area can be seen in **Figure 13**. This will appear if the user selects this plane from the data cube.

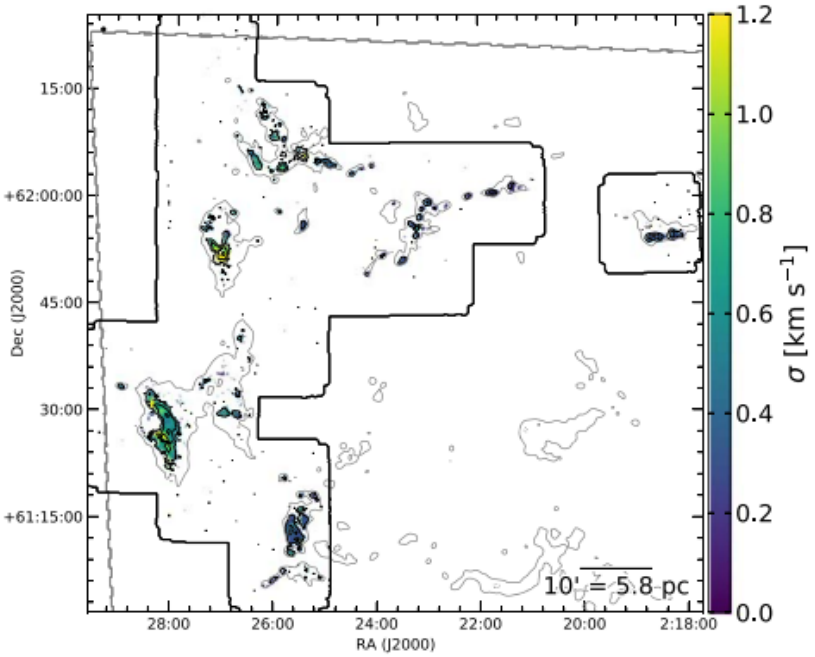


***Figure 12**: Velocity dispersion of NH_3 (1,1) obtained from line fitting using (1,1) and (2,2) for W3. Image taken from the KEYSTONE survey [7].*

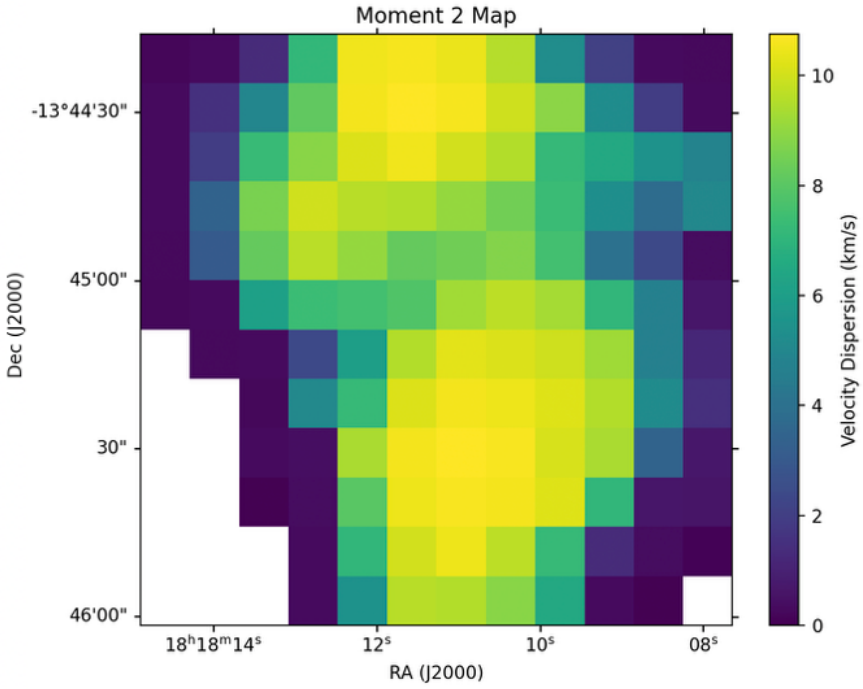


***Figure 13**: Moment 2 map for [KDR2019] M16 5 (RA 18:18:20.4 and Dec -13:46:16) querying around a circular region of 60 arcseconds.*

3.2.2 Temperature

The header in the KEYSTONE files contains BUNIT = " / Tmb where T_{mb} is the main-beam temperature and BUNIT describes the physical units of the quantities in the array [4], [40]. The data in the cube is already in terms of temperature (K) and from BUNIT the temperature contained within the file is the main-beam temperature [31]. The KEYSTONE survey used the GBT and so the source brightness temperature (T_s) can be approximated by the main-mean temperature (T_{mb}) for sources larger than the telescope beam [40]. Therefore, the data from the cube in terms of Kelvin is plotted to visualize the brightness temperature of that region. This can be seen in **Figure 14**.

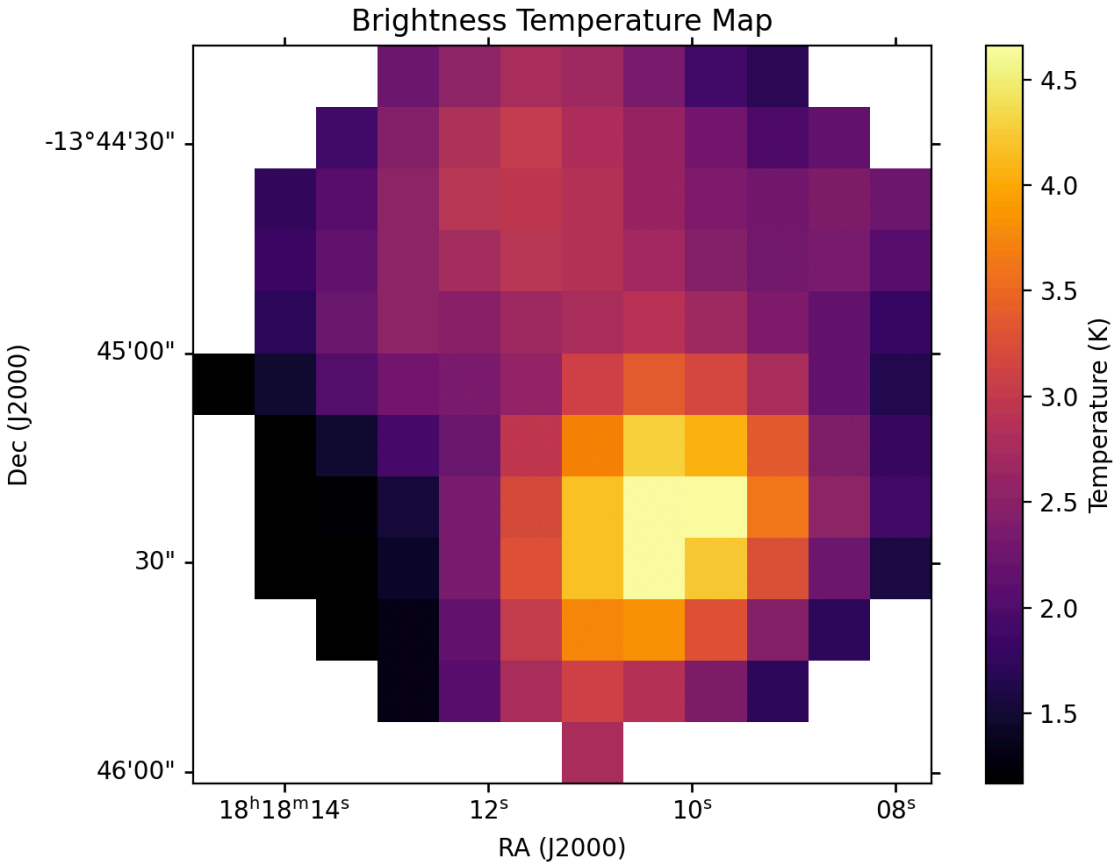


***Figure 14**: Intensity temperature (T_{mb}) [K] for [KDR2019] M16 5 (RA 18:18:20.4 and Dec -13:46:16) querying around a circular region of 60 arcseconds.*

3.2.3 Mean and Median over Area

The mean and median over the area are calculated and displayed using the intensity. The function takes the intensity from Moment 0 and extracts each intensity. It then uses these

intensities to calculate either the selected mean (eq. 9) or median (eq. 10 and 11) before displaying the value to the user [41].

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i \tag{9}$$

$$\tilde{x}\,(odd) = x\frac{n+1}{2} \tag{10}$$

$$\tilde{x}\,(even) = \frac{1}{2}(x\frac{n}{2} + x\frac{n}{2} + 1) \tag{11}$$

An example can be seen for [KDR2019] M16 5 with a circular region of 30 arcseconds that has a median of 10.947 km/s and a mean of 11.411 km/s.

3.2.4 Average over Spectrum

The average over the spectrum in the spectral cube can be mapped using the velocity or frequency and the brightness temperature (T_b). The function for computing the average spectrum first establishes the shape of the cube. This is necessary since even though the KEYSTONE files are 3D cubes, the mapped CO in **Figure 15** is a 4D cube that needs to be converted to 3D before the maximum pixel values can be found [42]. The 3D spectral cube can be visualized as containing a line profile for each 2D pixel [43]. The program also attempts to remove any noise before using equation 9 to compute the average. The function is called when the user selects the average spectrum plot. However, the KEYSTONE files already contain the brightness temperature, which does not apply to every spectral cube. The mapped CO from TW Hya contains a spectral cube that instead contains flux density (Jy/beam) that can be converted using the Rayleigh-Jeans law (equation 12)

$$T_b = \frac{\lambda^2}{2k} I_v \tag{12}$$

where λ is the wavelength, T_b is the brightness temperature, k is the Boltzmann Constant (1.380649J/K), and I_v is the intensity [43], [44]. The intensity can be related to the flux density, S, in equation 13 to produce equation 14 where Ω is the beam solid angle [43], [44].

$$\frac{S}{\text{beam}} \cdot \frac{beam}{\Omega} = I\frac{beam}{\Omega} \tag{13}$$

$$T_b = \frac{\lambda^2}{2k\Omega} S \tag{14}$$

A plotted average spectrum can be seen in **Figure 15** that reveals a single Gaussian fit. It can be compared to the CO spectra also seen in **Figure 3**. The URL from Dataverse was added to the KEYSTONE URLs to test this data type. The spectrum in **Figure 15** maps CO data from TW Hydra, a protoplanetary disc approximately 60 pc away, as the disc has CO as its primary gas-phase reservoir [45],[46].

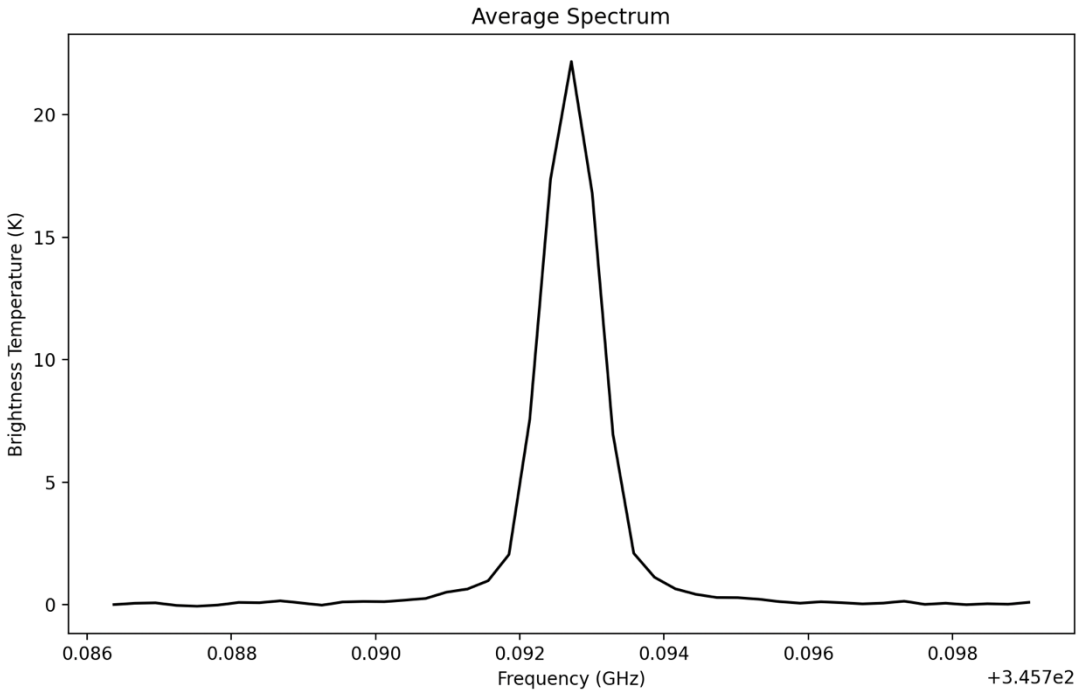


Figure 15*: Average spectrum plot for 13CO J=3-2 line cube from Dataverse in the dataset of [46]. A circular region of 30 arcseconds was selected.*

4 Conclusions

To conclude, a new astronomical search tool was designed in this thesis to query FITS files in the KFPA Examinations of Young Stellar Object Natal Environments (KEYSTONE) survey and other imaging surveys. These files are stored and located on the Canadian Advanced Network for Astronomical Research (CANFAR) platform. The new database, coined FitsQuery given its primary querying capabilities, was first proposed as a means of alleviating the need to manually download and search FITS files. The program instead offers a faster

means of querying by scanning through the files and examining whether the point of interest is located within the files or announcing that no file is found. In addition, FitsQuery offers various types of snapshot images of that region, such as Moment maps.

The first prompt in FitsQuery is to enter a target to query either by its object name or celestial coordinates. The program uses this input and converts it to pixel coordinates, using astropy, to locate a matching FITS file within the KEYSTONE survey. When a match is found, the program will stop its search and announce that a matching file has been located; otherwise, the program will continue to run before announcing that no files were found. The located file can now be read, and the program prints information, such as number of planes. This leads into the next prompt of selecting a spectral plane given the number of planes available. This is followed by additionally asking for a specific region whereby they can select either a circle or rectangle and its parameters. This information is used by the program in subsequent calculations. The final prompt is a menu containing different types of output options. It includes Moment maps, mean and median over the area, temperature, and average spectrum. After a type is selected, it displays the image or number and gives the option of saving the image. The options are looped so other types may be selected until the exit option is chosen.

As previously mentioned, FitsQuery was designed and programed around the KEYSTONE survey. However, the program should be able to be easily altered to instead query another dataset or survey by altering the code. This was tested and confirmed using ALMA and Dataverse by successfully locating files with known coordinates. Nevertheless, the subsequent snapshot images that accompany the program may still be tailored towards querying the KEYSTONE survey. Thus, it may only be programmed towards that specific data type and units. This criterion has the risk of not universally applying to all astronomical data available in other surveys, catalogues, and archives. Further recommendations and advancements of this database must include altering the output options to account for processing any astronomical data as much as possible. This could be achieved by standardizing the functions more and creating an algorithm to match the different types in the FITS header and then processing them accordingly. For instance, if CUNIT3 is in Hz or already in terms of km/s, it would change how the doppler shift is calculated. In the future, the program must include making the images and calculations more universal so that they can apply for a multitude of data types

and not just GMCs. Any alterations should be as minimal as possible to ensure that it is user friendly. Additional testing should be conducted using other data types to verify its adaptability. This will also allow for other databases and catalogues to easily be integrated into the code and allow for the user to more easily switch targets when querying. For instance, one potential of the program is to code multiple databases so that the user does not have to continually alter the code. Instead, the code would check one database and, if no match was found, it would continue to query another archive. There is much potential for this program and to further its ability to aid astronomers and physicists that should continue to be explored and implemented.

Word Count: 4980

Appendix A. Statement of Work

The work done in the fall term consisted of researching FITS files and understanding how to process astronomical data. By the end of the fall term FitsQuery was able to query the KEYSTONE files and output how many files matched the celestial coordinates. In the winter term, I added object names as a second initial search parameter, and all of the selection options. This included selecting a region, plane, and the output options. In addition, I tested if FitsQuery could query ALMA archives and Dataverse.

Appendix B. Alternative Catalogues

Alma

FitsQuery can target the ALMA archive by adding

```
from astroquery.alma import Alma
import sys
```

at the top of the code. The next step is to swap a section of the main code with one that is specific to querying in ALMA. This section in FitsQuery

```
with open(Fits_Files) as f:
       urls = [l.strip() for l in f if l.strip()]
coord = resolve_target()
result = find_matching_keystone(coord, urls)
if result is None:
print("No file contains these coordinates.")
exit()
url, hdr, pixel = result
```

is replaced with

```
radius=10
print("\nQuerying the ALMA archive...\n")
coord = resolve_target()
obs = Alma.query_region(coord, radius=radius*u.arcsec)
if len(obs) == 0:
       print("No file contains these coordinates.")
       sys.exit(0)
print(f"Found {len(obs)} observations")7
```

Dataverse

FitsQuery can also target Dataverse by replacing the KEYSTONE text file with the URL for Dataverse.

```
Fits_Files = "https://dataverse.harvard.edu/api/search"
```

The next step is to swap a section of the main code with another that is specific to querying Dataverse. This section in FitsQuery

```
with open(Fits_Files) as f:
       urls = [l.strip() for l in f if l.strip()]
coord = resolve_target()
result = find_matching_keystone(coord, urls)
if result is None:
print("No file contains these coordinates.")
exit()
url, hdr, pixel = result
```

is replaced with

```
print("\nQuerying Dataverse...\n")
coord = resolve_target()
params = {
       "q": "astronomy",
       "type": "file" }
r = requests.get(Fits_Files, params=params)
r.raise_for_status()
data = r.json()
files = data.get("data", {}).get("items", [])
if len(files) == 0:
       print("No file contains these coordinates.")
       sys.exit(0
print(f"Found {len(files)} files")
```

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